

GCP VPCs (1 of 2)

- **Virtual Firewall**
 - GCP firewall works at the **network level** to control **IPv4** traffic into and out of the VPC
 - Rules can target specific protocols, ports, sources, or destinations
 - Firewall rules are applied by **priority** first, then by how specifically they match the traffic
- **VPC Flow Logs**
 - VPC flow logs are enabled at the subnet level
 - They track TCP and UDP network traffic
- **Packet Mirroring**

Securing Hybrid and Multi-Cloud

- Cloud security tools must **auto-scale**, while on-prem security tools need deep access inside the network
- Many cloud security best practices also apply to hybrid and multi-cloud environments
- A common tool to connect on-prem and cloud networks is a **CASB (cloud access security broker)** solution
- A **CASB** detects cloud applications and resources and monitors them using organization security policies

Compute Security

- Cloud compute resources usually include instances and applications
- This section explains security techniques used to protect instances from inside the instance

Device Hardening

- When installing an OS on a server, you should create a **hardened baseline**
It documents all configuration changes needed to secure the system and the OS
- Consider these security techniques:
 - **Plan subnet placement (public/private?)**
 - **Disable unnecessary ports and services**
 - **Enforce account management policies**
 - **Install antivirus and anti-malware software and keep it updated**
 - **Install host-based or software firewalls and IDS or IPS**
 - **Create single-function resources**

Application Security

- Application security applies to SaaS applications and to PaaS components used to build applications
- A **WAF** is a special firewall used to protect SaaS applications
 - A WAF can inspect, monitor, and filter incoming web traffic to protect the application
- An **ADC** is a powerful load balancer that can also:
 - Filter traffic in a more detailed way
 - Handle encryption to reduce server load
 - Support traffic shaping and monitoring

ADC (Application Delivery Controller)

WAF (Web Application Firewall)

Normal firewall:

- IP
- Port
- Protocol bakar

WAF ise şuna bakar:

- HTTP request içeriği
- URL
- Header
- Body
- Parametreler

Data Security

- Security controls must move together with the data
- Data exists in three states:
 - **At rest** means data stored in file, block, or object storage.
 - **In use** means data is temporarily stored in memory or processed by a system.
 - **In motion** means data is being transferred between locations.
- **DLP(Data Loss Prevention)** solutions protect data in all three states. They focus on preventing data loss using tools like CASBs and secure web gateways.

Encrypting Data (1 of 3)

- **Encryption** uses an algorithm to change data into a form that can only be read by reversing the algorithm
- Cloud data encryption can be done by the **data owner**, the **cloud service provider**, or a **third-party encryption service**
- To deploy encryption in the cloud, these tasks must be completed:
 - Understand and use the encryption features and methods of the CSP
 - Decide where encryption keys are stored
 - Consider the costs of each encryption option

Encrypting Data (2 of 3)

- **Encryption Techniques**
 - Encryption methods are based on the **CIA triad** to protect data at rest, in use, and in motion:
 - **Confidentiality** means data can only be accessed by the receiver.
 - **Integrity** means data is not changed during transfer or storage
 - **Availability** means data is accessible when needed
 - Encryption protocols use a mathematical code called a **cipher**
A cipher scrambles data and can only be reversed to read the original data

Encrypting Data (3 of 3)

- **Encryption Techniques**
 - Strong encryption uses a long and complex **key** to scramble data and create encrypted data called **ciphertext**
 - Encryption keys are grouped into two types:
 - **Private key encryption** uses one shared key that both sender and receiver know
 - **Public key encryption** uses two keys
Data is encrypted with a public key and decrypted with a related private key that can be shared through a trusted source

Securing Protocols (1 of 4)

- **SSL and TLS** are encryption protocols used to secure communication over the Internet
- Each SSL or TLS connection creates a unique session
 - The session is created using a **handshake process**
The client and server identify each other and agree on how to securely exchange data
- **OpenSSL** is a command-line tool used to troubleshoot SSL and TLS encryption and certificates
 - The **s_client** command can test secure connections, check security settings, and manage certificates

Securing Protocols (3 of 4)

- **DNS Security**
 - DNS messages are **unencrypted** and **unauthenticated** by default
 - The following technologies help secure DNS traffic:
 - **Authentication** uses **DNSSEC** (DNS Security Extension) to verify DNS records with public key cryptography.
 - **Encryption** uses **DoT** (DNS over TLS) and **DoH** (DNS over HTTPS) to encrypt DNS queries.

Securing Protocols (4 of 4)

- **NTP(Network Time Protocol) Security**
 - NTP is used to synchronize time across systems in a network
 - NTP does not provide authentication by default
It can be secured using **NTS**, which uses TLS to authenticate time servers

Troubleshooting Cloud Security

- Security issues can create serious risks for a company
- This section explains common security troubleshooting scenarios in cloud environments

Security Misconfigurations (1 of 3)

- Cloud security depends on correct configurations and regular maintenance
- If traffic does not reach its destination, the cause is often:
 - A wrong or misapplied security policy or ACL in a firewall or NSG
 - A misconfigured or failed security appliance
 - Incorrectly managed micro-segmentation
- Cloud platforms provide tools to check security best practices in configurations
 - AWS provides Trusted Advisor

Security Misconfigurations (3 of 3)

- Common cloud security weak points include:
 - Unencrypted data and unencrypted communication
 - Weak or outdated security technologies
 - Incorrect hardening settings
 - Security device failure
 - Insufficient security controls and processes
 - Unauthorized physical access

Key Management

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- To use public key encryption in the cloud, private keys must be stored on-prem or in the cloud using a key management service
 - Key management issues you may face include:
 - Unavailable keys
 - Failed connections
 - Encryption operation failures
 - To fix key storage problems, make sure required keys are set as available and not deleted
 - Also check the configuration of resources that try to access the keys